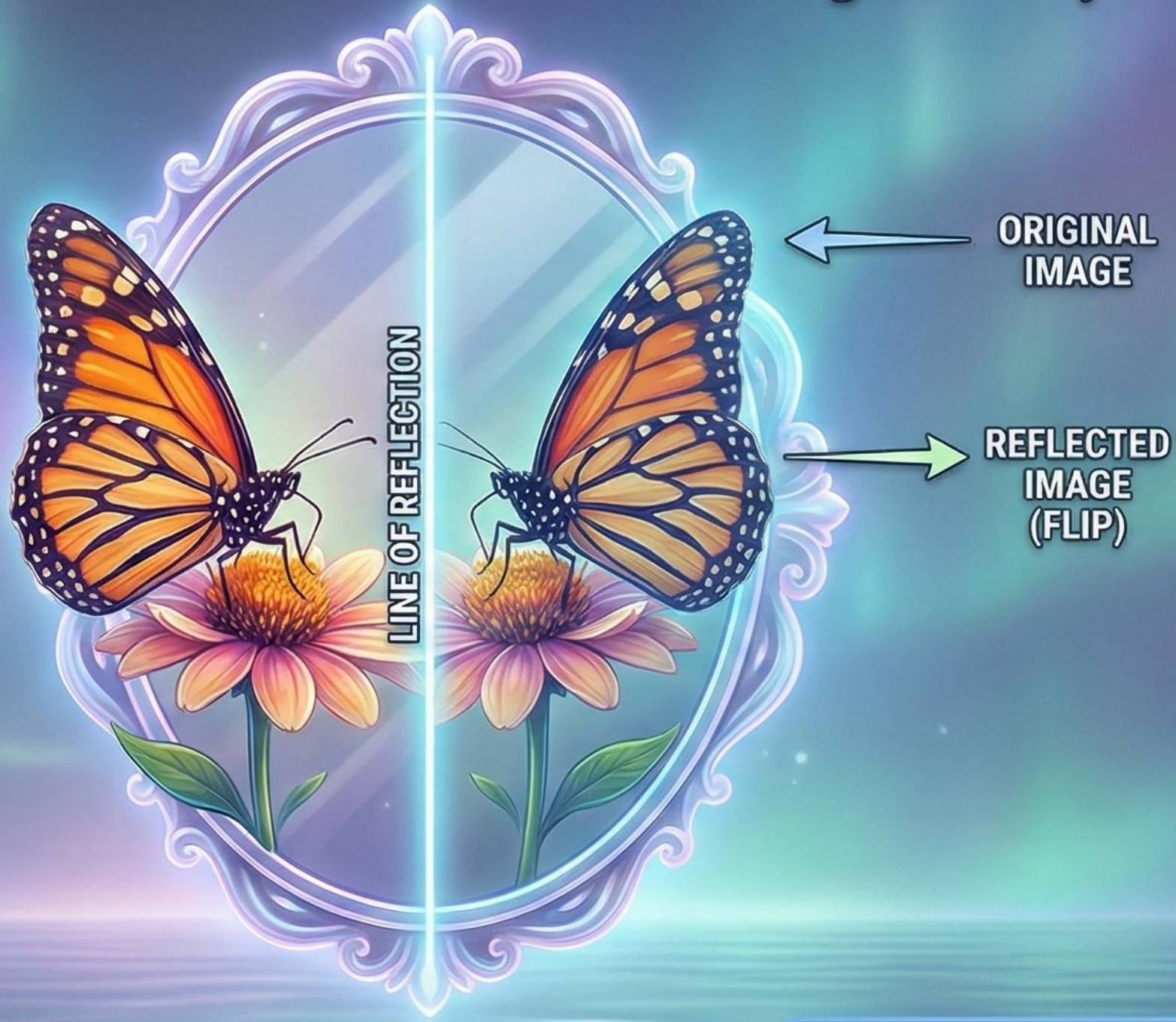
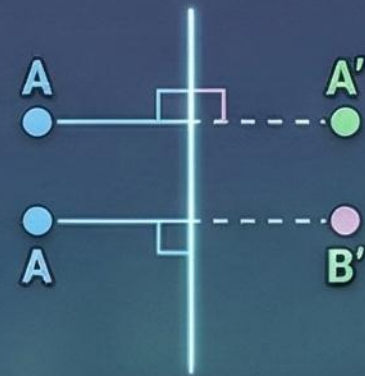


THE GEOMETRY OF FLIP: REFLECTION & SYMMETRY

— ≈ A Rigid Transformation Across a Line ≈ —



DISTANCE PRESERVED



Every point and its reflection are the **SAME DISTANCE** from the line of reflection.

REVERSAL OF ORIENTATION



The image is a **MIRROR REVERSE**. Left becomes Right, and Right becomes Left.



CREATES SYMMETRY



When a shape and its reflection form a single, balanced figure, it has **BILATERAL SYMMETRY**.

REAL-WORLD EXAMPLES

NATURE



Lake Reflection

ART & DESIGN



Kaleidoscope

TECHNOLOGY

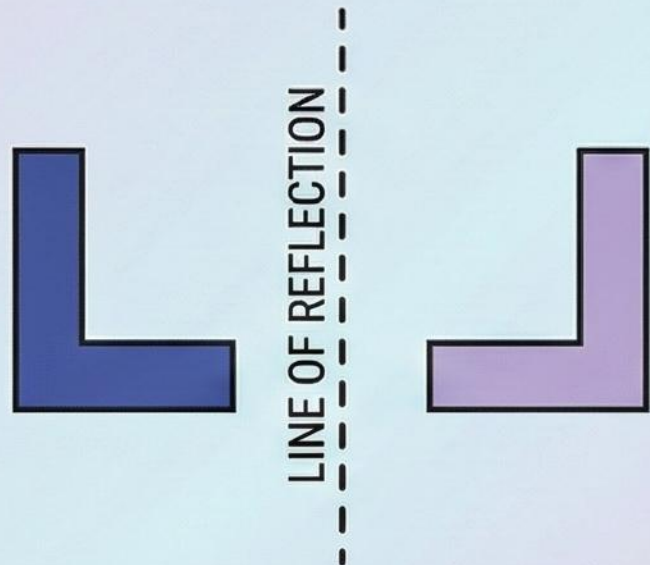


Rearview Mirror

Flips are fundamental transformations that maintain size and shape, altering only orientation and handedness, a principle seen throughout nature and design.

GEOMETRY OF MOVEMENT: FLIP, TURN, & SLIDE

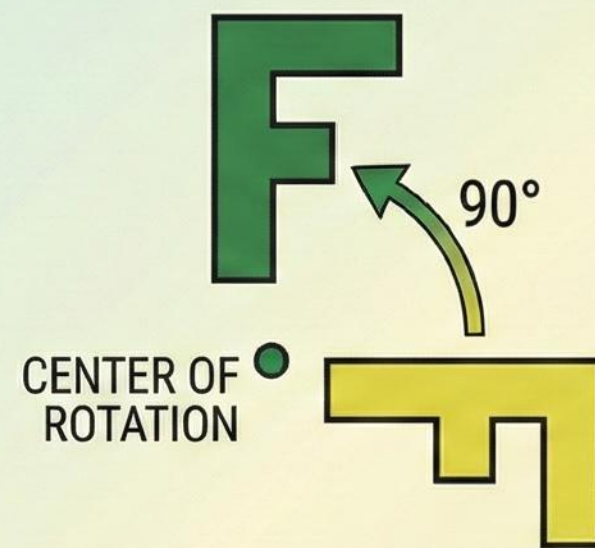
1. FLIP (REFLECTION)



A transformation that creates a mirror image across a line. The shape is reversed.



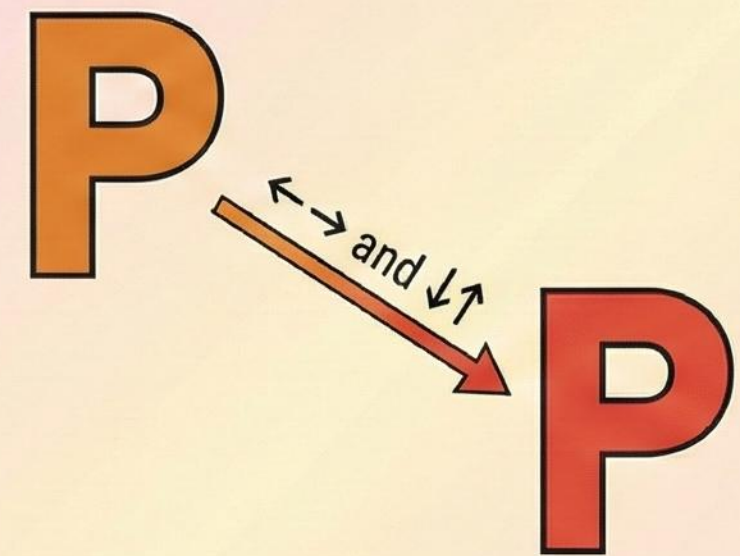
2. TURN (ROTATION)



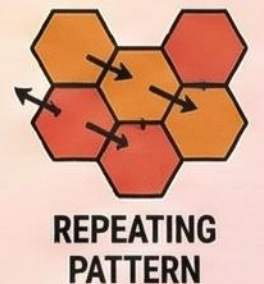
A transformation that turns a shape around a fixed point. The orientation changes.



3. SLIDE (TRANSLATION)



A transformation that moves every point of a shape the same distance in the same direction. The shape's orientation remains the same.



TRANSFORMATIONS IN EVERYDAY LIFE

These are **RIGID TRANSFORMATIONS**, meaning they preserve the shape and size of the object, only changing its position or orientation.



SLIDE (TRANSLATION) IN GEOMETRY

A **RIGID TRANSFORMATION** that moves every point of a figure by the **SAME DISTANCE** in the **SAME DIRECTION**.

KEY PROPERTIES:

Original Figure



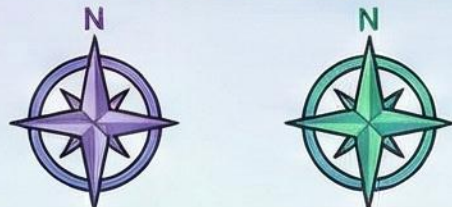
Slide Vector



Translated Figure



NO ROTATION



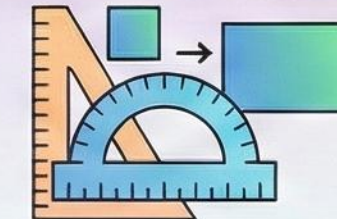
Orientation Preserved

NO REFLECTION



Image is **NOT** flipped

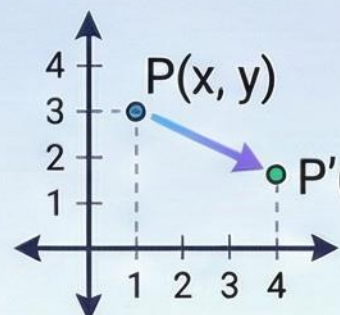
SAME SIZE & SHAPE



Lengths & Angles
Remain Constant

ISOMETRY

COORDINATE RULE



Shift by (a, b):
 $(x, y) \rightarrow (x+a, y+b)$

REAL-WORLD EXAMPLES



Everyday movement without twisting or flipping.

THE BEAUTY OF SYMMETRY: BALANCE & HARMONY IN FORM

EXPLORING REFLECTIONAL, ROTATIONAL, & TRANSLATIONAL SYMMETRY

REFLECTIONAL (BILATERAL) SYMMETRY

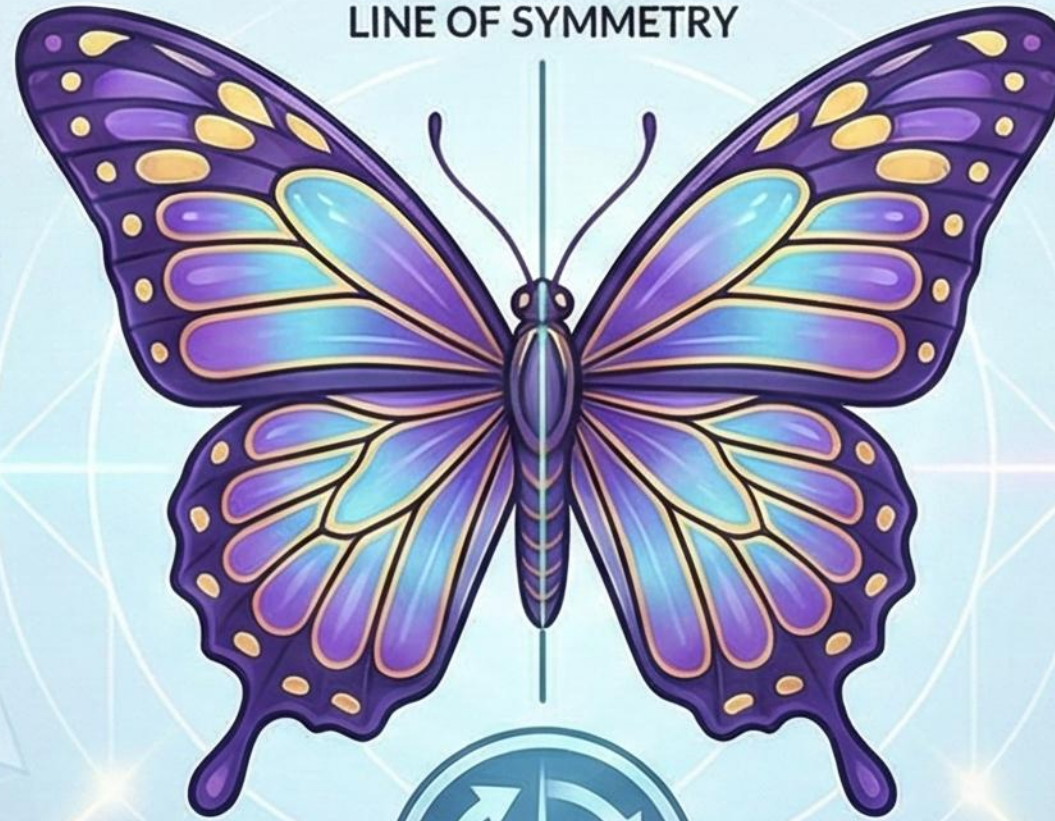


Mirror Image across a line.
One half is a reflection of the other.



Found in: Faces, Animals, Architecture

LINE OF SYMMETRY



CENTER OF ROTATION



ROTATIONAL SYMMETRY



Shape looks the same
after rotation by a partial turn.



Found in: Flowers, Wheels, Logos.
Order of Symmetry: Number of times it matches in 360°.

SYMMETRY IN OUR WORLD



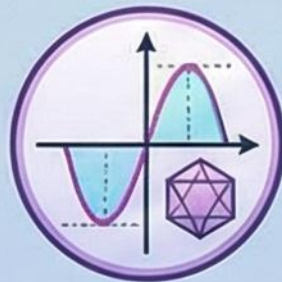
NATURE

Growth & Efficiency.



ART & DESIGN

Aesthetics & Balance.



MATHEMATICS

Structure & Logic.



TRANSLATIONAL SYMMETRY

Repetition of a pattern by
sliding it along a straight line.

Found in: Patterns, Tiling, Textiles, Crystals.



Found in: Patterns, Tiling, Textiles, Crystals.



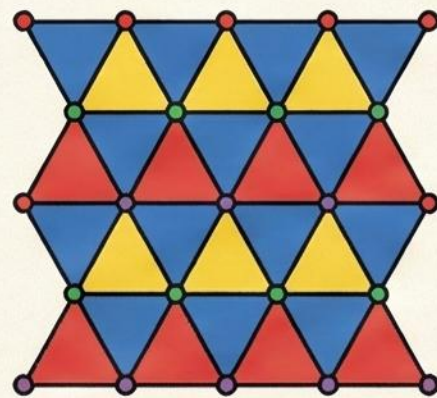
Symmetry is a fundamental principle of order and beauty,
bringing a sense of calm and proportion to the universe.



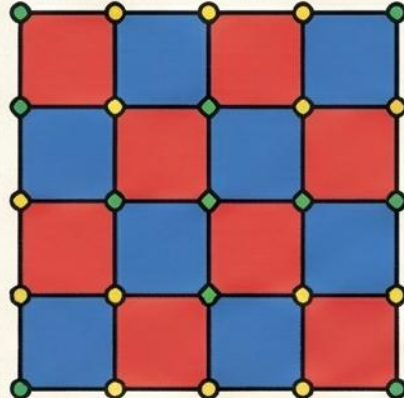
The Art & Geometry of Tessellating Patterns

A plane tiled with shapes, no gaps, no overlaps, repeating infinitely.

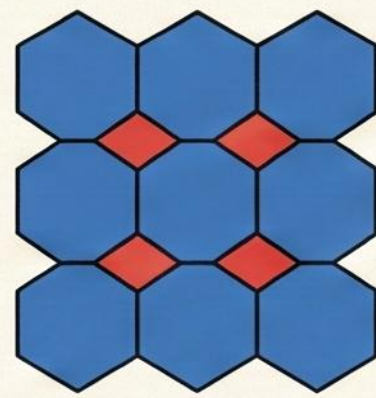
Regular Tessellations



3.3.3.3.3.3



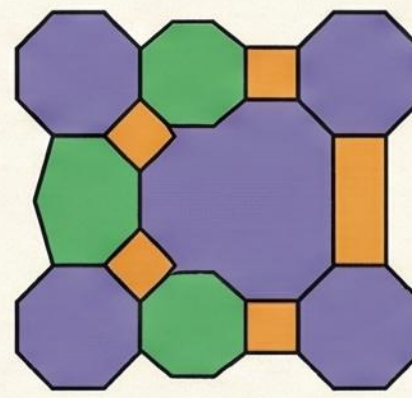
4.4.4.4



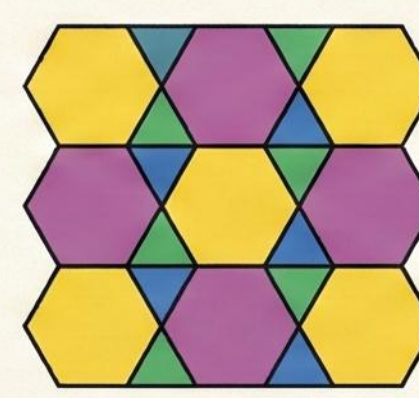
6.6.6

Only one type of regular polygon;
vertex arrangement is identical.

Semi-Regular Tessellations



4.8.8



3.4.6.4

Two or more regular polygons;
vertex arrangement is identical.

Irregular Tessellations



Made of non-regular polygons. No strict rules for
vertex arrangement, but must tile without gaps.

Special Cases & Real-World Examples

Aperiodic Tessellation (Penrose Tiling)

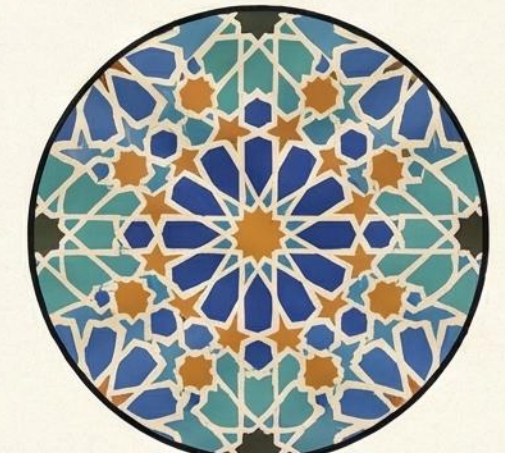


Does not repeat periodically. Infinite pattern with no translational symmetry.

Nature & Architecture



Honeycomb
(Natural hexagon tiling)



Islamic Zellige
(Artistic tiling)

THE GEOMETRY OF A TURN (ROTATION)

DIRECTION MATTERS



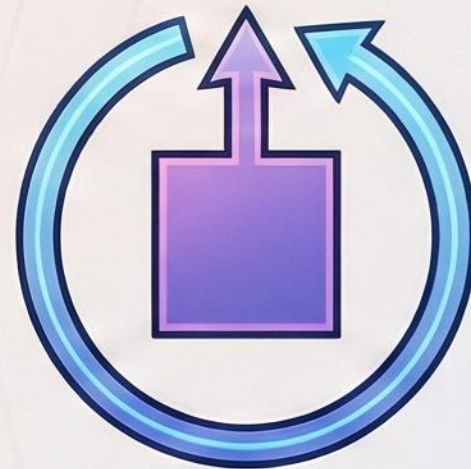
CLOCKWISE
(CW)



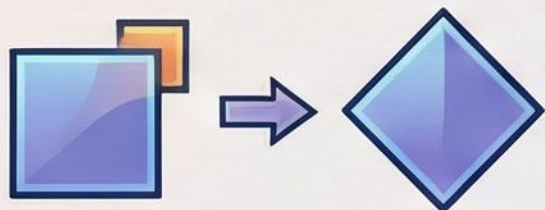
COUNTER-
CLOCKWISE
(CCW)

FULL TURN (360°)

1 Full Turn = 360 Degrees



RIGID TRANSFORMATION



Shape & Size remain **CONSTANT**

QUARTER TURN (90°)



1/4 Turn = 90 Degrees



CENTER OF
ROTATION
(FIXED POINT)



THREE-QUARTER TURN (270°)

3/4 Turn = 270 Degrees

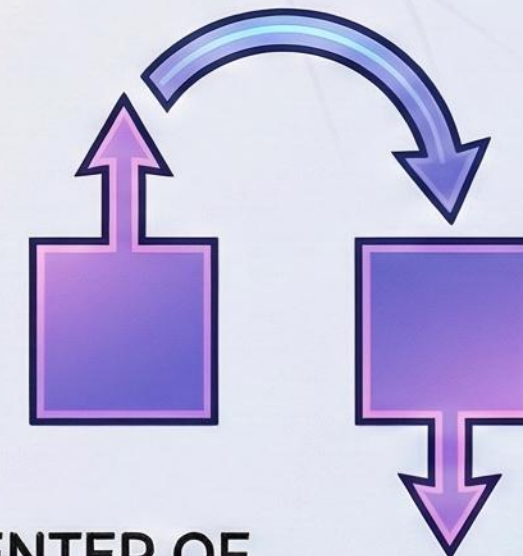
MEASURING TURNS



Degrees (°) or Fractions

HALF TURN (180°)

1/2 Turn = 180 Degrees



REAL-WORLD EXAMPLES

